CHAPTER # 5

PROBABILITY

Probability

The measure of uncertainty is called probability.

Experiment

The term experiment mean a planned activity or process whose result yield a set of data.

Trial

A single performance of an experiment is called trial.

Outcome

The result obtained from an experiment or a trial is called outcome.

Random Experiment

A experiment which produced different results even through it is repeated a large number of time under essentially similar conditions is called random experiment.

For example, Tossing of fair coin, the throwing of balanced dice, Drawing of a card from a well shuffled deck of 52 playing cards.

Sample space

A set consisting of all possible outcomes that can results from a random experiment is called sample space.

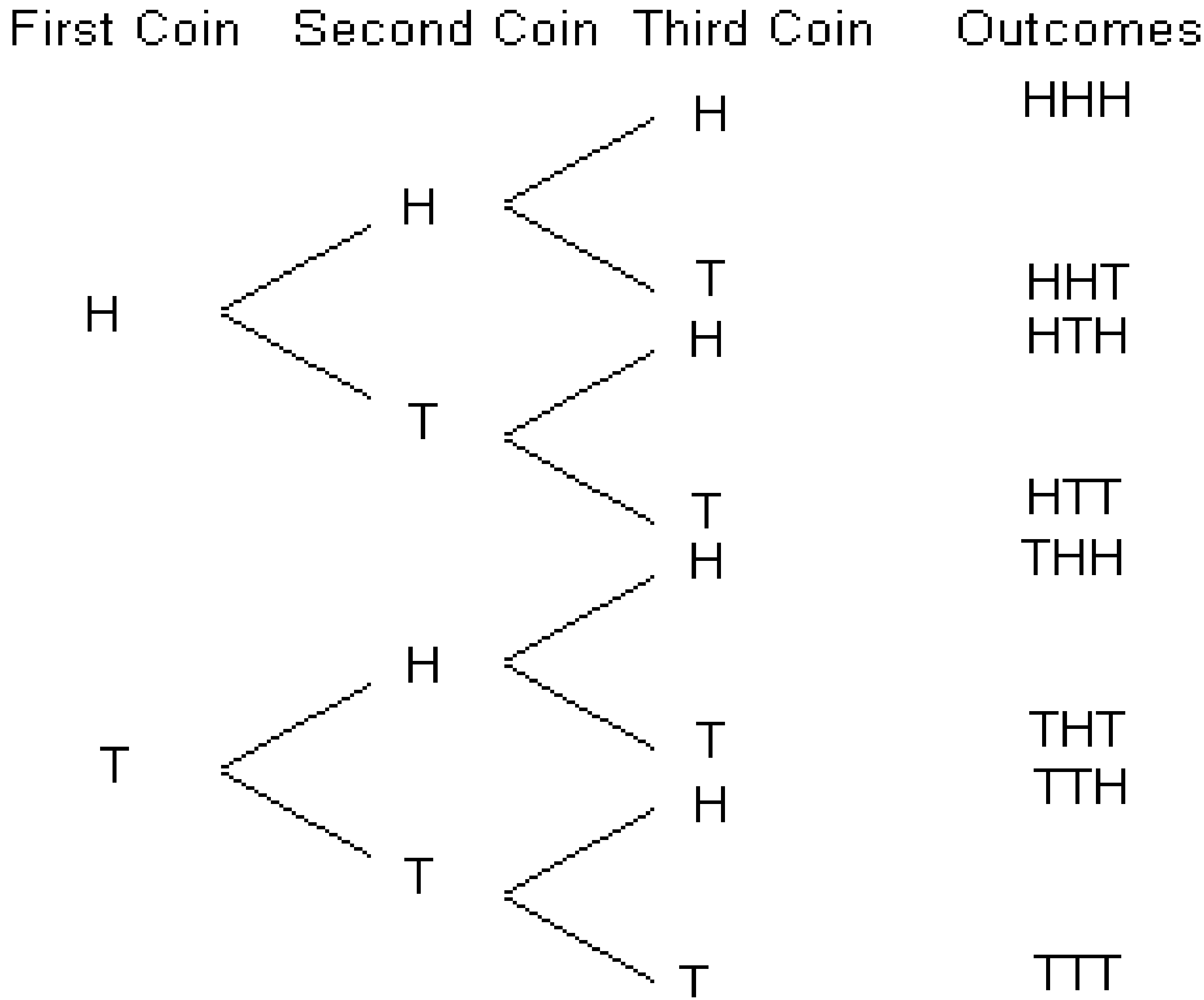
For example.

1. A fair coin is toss the total sample space is $(2^1 = 2)$

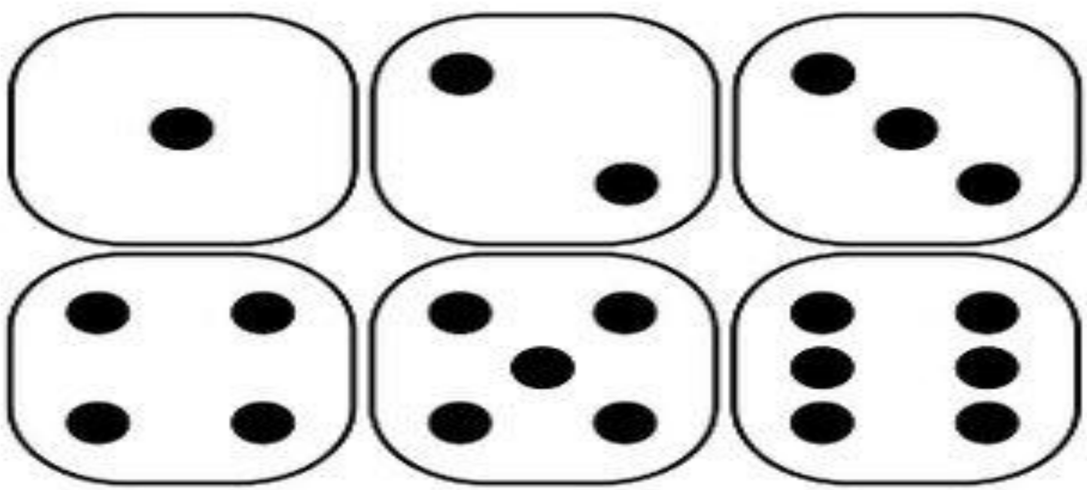
$S = \{H, T\}$

2. If we toss two coin the total sample space is $(2^2 = 4)$

$S = \{HH, HT, TH, TT\}$


First Coin
Second Coin
Third Coin
Outcomes
H
T
H
T
H
T
H
T
H
T
H
T
H
T
HHH
HHT
HTH
HTT
THH
THT
TTH
TTT

4. If we throwing one dice then total sample space $6^1 = 6$



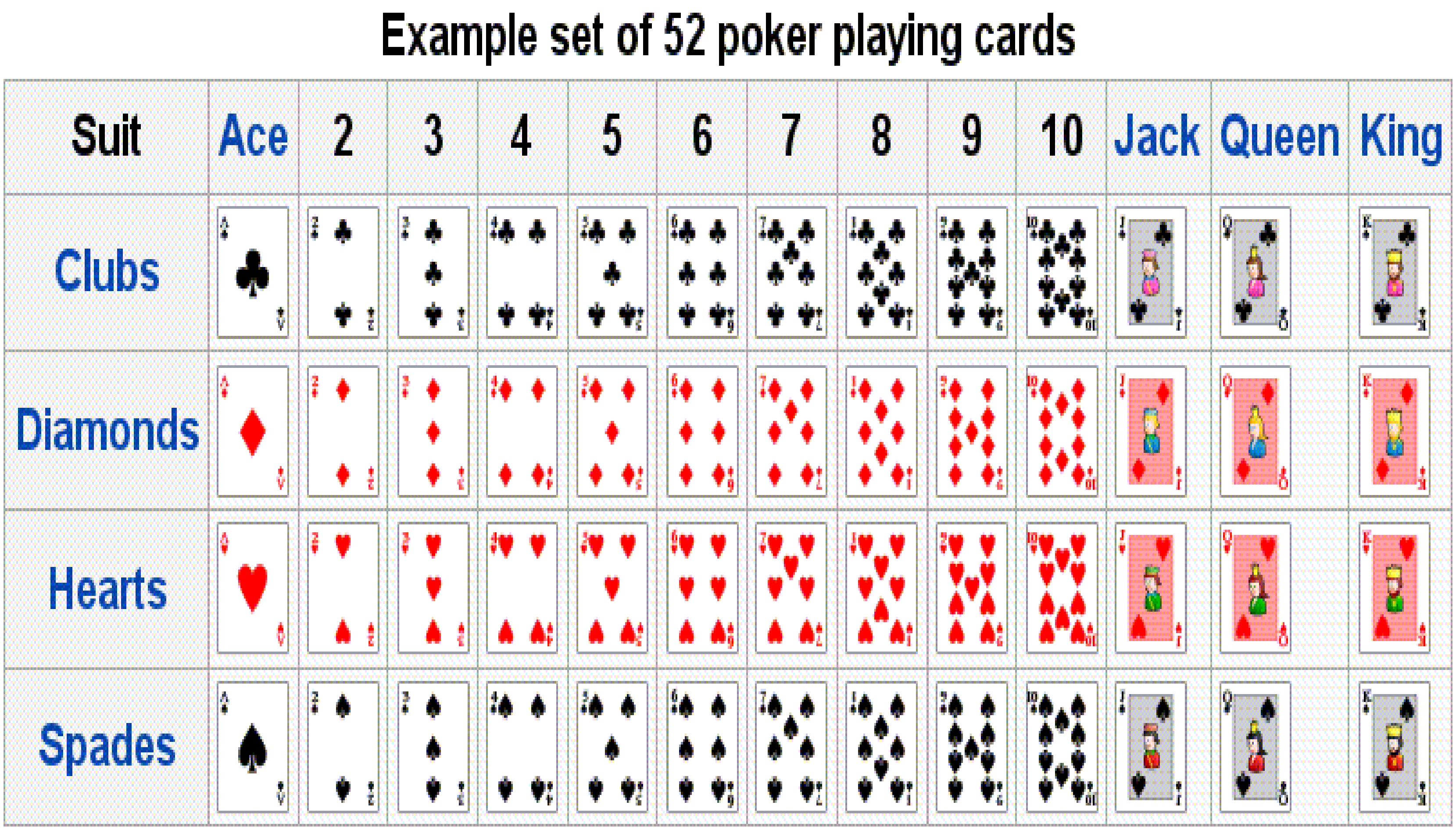
$$\boldsymbol{S = \{1, 2, 3, 4, 5, 6\}}$$

5. If we throwing two dice then total sample space $6^2 = 36$

		second dice					
		1	2	3	4	5	6
first dice	1	(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
	2	(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
	3	(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
	4	(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
	5	(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
	6	(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$$S = \begin{bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{bmatrix}$$

6. Drawing of a card from a well shuffled deck of 52 playing cards



How we can find Probability of an event A

If an event A is defined in a sample space S, then

probability will be equal to

$$P(A) = \frac{Number\ of\ favourable\ outcome}{Total\ number\ of\ possible\ outcome}$$

$$P(A) = \frac{n(A)}{n(S)}$$

Example 1: Two coins is toss, what is the probability that two head appears at a same time.

Solution: the sample space for two coins is

$$S = \{HH, HT, TH, TT\}$$

$$n(S) = 4$$

Two head appears at same time events is

$$A = \{HH\}$$

$$n(A) = 1$$

Now the probability of two head is

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{1}{4}$$

Example 2: Three coins is tossed, what is the probability that

(1). At least one head appears

(2). Head on the first coin

Solution: The sample space for tossing three coins is

$S = \{HHH, HTH, THH, TTH, HHT, HTT, THT, TTT\}$

$$n(S) = 8$$

(1). At least one head appear events are

$$A = \{HHH, HTH, THH, TTH, HHT, HTT, THT\}$$

$$n(A) = 7$$

Now the probability of at least one head is

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{7}{8}$$

2). Head on the first coin

$$B = \{HHH, HTH, HHT, HTT\}$$

$$n(B) = 4$$

Now the probability is

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{4}{8}$$

$$P(B) = \frac{1}{2}$$

Example 3: If two fair dice are thrown, what is the probability of getting

(1). A double six appears

(2). The sum of 8 or more dots

(3). The two dice have same face appears

(4). The sum is odd number

Solution: The sample space for thrown two dice is

$$S = \begin{bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{bmatrix}$$

$$n(S) = 36$$

(1). A double six appears

$$A = \{(6,6)\}$$
$$n(A) = 1$$

Now the probability of double six is

$$P(A) = \frac{n(A)}{n(S)}$$
$$P(A) = \frac{1}{36}$$

(2). The sum of 8 or more dots

$$B = \begin{Bmatrix} (2,6),(3,5),(3,6),(4,4),(4,5),(4,6),(5,3),(5,4), \\ (5,5),(5,6),(6,2),(6,3),(6,4),(6,5),(6,6) \end{Bmatrix}$$
$$n(B) = 15$$

Now the probability of sum 8 or more is

$$P(B) = \frac{n(B)}{n(S)}$$
$$P(B) = \frac{15}{36} = \frac{5}{12}$$

(3). The two dice have same face appears

$$C = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$$

$$n(C) = 6$$

Now the probability same face is

$$P(C) = \frac{n(C)}{n(S)}$$

$$P(C) = \frac{6}{36} = \frac{1}{6}$$

(4). The sum is odd number $D =$

$$\begin{Bmatrix} (1,2),(1,4),(1,6),(2,1),(2,3),(2,5),(3,2),(3,4),(3,6), \\ (4,1),(4,3),(4,5),(5,2),(5,4),(5,6),(6,1),(6,3),(6,5) \end{Bmatrix}$$

$$n(D) = 18$$

Now the probability sum is odd

$$P(D) = \frac{n(D)}{n(S)}$$

$$P(D) = \frac{18}{36} = \frac{1}{2}$$

Example 4: If a card is drawn from an ordinary deck of 52 playing cards, find the probability of

(1). The Card is Red card

(2). The card is diamond card

(3). The face card

(4). King card

(5). The number card

(6). The Ace card

Solution: The sample space of playing card

$$n(S) = 52$$

(1). The Card is Red card

the total number of Red card is 26

$$n(A) = 26$$

Now the probability of red card is

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{26}{52} = \frac{1}{2}$$

(2). The card is diamond card

the total number of Diamond card is 13

$$n(B) = 13$$

Now the probability of diamond card is

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{13}{52} = \frac{1}{4}$$

(3). The face card

the total number of face card is 12

$$n(C) = 12$$

Now the probability of face card is

$$P(C) = \frac{n(C)}{n(S)}$$

$$P(C) = \frac{12}{52} = \frac{3}{13}$$

(4). King card

the total number of king card is 4

$$n(D) = 4$$

Now the probability of king card is

$$P(D) = \frac{n(D)}{n(S)}$$

$$P(D) = \frac{4}{52} = \frac{1}{13}$$

(5). The number card

the total number of Number card is 36

$$n(E) = 36$$

Now the probability of Number card is

$$P(E) = \frac{n(E)}{n(S)}$$

$$P(E) = \frac{36}{52} = \frac{9}{13}$$

(6). The Ace card

the total number of Ace card is 4

$$n(F) = 4$$

Now the probability Ace card is

$$P(F) = \frac{n(F)}{n(S)}$$

$$P(F) = \frac{4}{52} = \frac{1}{13}$$

Events

An events is an individual outcome or any number of outcomes of a random experiment or a trial. Any subset of a sample space S of the experiment is called events.

An events that contains exactly one sample points is called simple events, occurrence of 6 when a dice is thrown is simple events.

An events contains more than one sample point is called compound events, occurrence of sum 10 when two dice is thrown is compound events.

Mutually Exclusive Events

Two events A & B of single experiments are said to be mutually exclusive events if and only if they cannot both occur at the same time. That is they have no points in common. For example when we toss a coin we get either a head or a tail but not both, the two events head and tail are mutually exclusive events.

If $A = \{1,3,5\}$ and $B = \{2,4,6\}$ then A & B are said to mutually exclusive events.

Exhaustive events

Event are said to be collectively exhaustive, when the union of mutually exclusive events is the entire sample space. Thus in our coin tossing experiment head and tail are collectively exhaustive set of events.

Equally likely events

Two events A & B are said to equally likely events, when one event is as likely to occur as the other. When a fair coin is tossed, the head is as likely to appear as a tail.

Factorial

The product of an integer and all the integer below it.

For example

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$5! = 120$$

Rule of Permutation

A permutation is any ordered subset from a set of n distinct objects. The number of permutation of r objects selected in a definite order from n distinct objects. It is denoted as P_r^n.

$$P_r^n = \frac{n!}{(n-r)!}$$

For example, Let $\{A, B, C\}$ given three numbers, how many number can be made be taking 2 number, order matters?
There are 6 possible outcome such as, $\{AB, BA, AC, CA, BC, CB\}$
Using Permutation

$$P_r^n = \frac{n!}{(n-r)!}$$

$$P_2^3 = \frac{3!}{(3-2)!} = \frac{3!}{1!}$$

$$P_2^3 = \frac{3 \times 2 \times 1}{1}$$

$$P_2^3 = 6$$

Example 1. A club consist of four members. How many sample points are in the sample space, when three officers, president, secretary and treasurer are to be chosen?

Solution: here order is given we used the permutation

$$P_r^n = \frac{n!}{(n-r)!}$$

$$P_3^4 = \frac{4!}{(4-3)!}$$

$$P_3^4 = \frac{4!}{1!}$$

$$\therefore 4! = 4 \times 3 \times 2 \times 1 = 24$$

$$P_3^4 = 24$$

Rule of Combination

A combination is any subset of r objects, selected without regard to their order, from a set of n distinct objects. It is denoted as C_r^n.

$$C_r^n = \frac{n!}{r!\,(n-r)!}$$

Let $\{A, B, C\}$ given three numbers, how many number can be made be taking 2 number, order dose not matters?
There are 3 possible outcome such as, $\{AB = BA, AC = CA, BC = CB\}$
using combination

$$C_r^n = \frac{n!}{r!\,(n-r)!}$$

$$C_2^3 = \frac{3!}{2!\,(3-2)!} = \frac{3!}{2! \times 1!}$$

$$C_2^3 = \frac{3 \times 2 \times 1}{2 \times 1 \times 1}$$

$$C_2^3 = 3$$

Example 2. A three person committee is to be formed from a list of four persons. How many sample points are associated with the experiment?

solution: using combination because no order are given here

$$C_r^n = \frac{n!}{r!\,(n-r)!}$$

$$C_3^4 = \frac{4!}{3!\,(4-3)!} = \frac{4!}{3! \times 1!}$$

$$C_3^4 = \frac{4 \times 3 \times 2 \times 1}{3 \times 2 \times 1 \times 1}$$

$$C_3^4 = 4$$

Laws of Probability

Here, we discuss two laws of probability

(1). Addition law

(2). Multiplication law

(1). Addition law

If in two events used “or“ that is A or B, then we used Addition law.

If two events are mutually exclusive event then we used formula for addition law is

$$P(A \cup B) = P(A) + P(B)$$

If two events are not mutually exclusive event then we used formula for addition law is

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Example 1: A pair of dice are thrown, find the probability of getting a total of either 5 or 11.

solution: The sample space of two dice are

$$S = \begin{bmatrix} (1,1) & (1,2) & (1,3) & (1,4) & (1,5) & (1,6) \\ (2,1) & (2,2) & (2,3) & (2,4) & (2,5) & (2,6) \\ (3,1) & (3,2) & (3,3) & (3,4) & (3,5) & (3,6) \\ (4,1) & (4,2) & (4,3) & (4,4) & (4,5) & (4,6) \\ (5,1) & (5,2) & (5,3) & (5,4) & (5,5) & (5,6) \\ (6,1) & (6,2) & (6,3) & (6,4) & (6,5) & (6,6) \end{bmatrix}$$

$$n(S) = 36$$

let A be the event that total 5 occur

$$A = \{(1,4), (2,3), (3,2), (4,1)\}$$

$$n(A) = 4$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{36}$$

let B be the event that total 11 occur

$$B = \{(5,6), (6,5)\}$$

$$n(B) = 2$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{2}{36}$$

The event A & B are mutually exclusive event so

$$P(A \cup B) = P(A) + P(B)$$

$$P(A \cup B) = \frac{4}{36} + \frac{2}{36} = \frac{6}{36}$$

$$P(A \cup B) = \frac{1}{6}$$

Example 2: If one card is selected at random from a deck of 52 playing cards, what is the probability that the card is club or face card?

Solution: The sample space of total card is

$$n(S) = 52$$

Let A denote the club card

$$A = \{A_c, 2,3,4,5,6,7,8,9,10, K_c, Q_c, J_c\}$$
$$n(A) = 13$$
$$P(A) = \frac{n(A)}{n(S)} = \frac{13}{52}$$

Let B denote the face card

$$B = \{K_c, Q_c, J_c, K_s, Q_s, J_s, K_h, Q_h, J_h, K_d, Q_d, J_d\}$$
$$n(B) = 12$$
$$P(B) = \frac{n(B)}{n(S)} = \frac{12}{52}$$

here, A & B are not mutually exclusive event

$$A \cap B = \{K_c, Q_c, J_c\}$$
$$n(A \cap B) = 3$$
$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{3}{52}$$

because A & B are not mutually exclusive event so we used

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = \frac{13}{52} + \frac{12}{52} - \frac{3}{52} = \frac{22}{52}$$

$$P(A \cup B) = \frac{11}{26}$$

Example 3: An integer is chosen at random from the first 200 positive integer, what is the probability that the integer chosen is divisible by 6 or by 8?

Solution: The sample space of first 200 positive integers is

$$S = \{1,2,3,4,5,\ldots,198,199,200\}$$

$$n(S) = 200$$

let A is the event that divisible by 6

$$A = \begin{Bmatrix} 6,12,18,24,30,36,42,48,54,60,66,72,78,84,90,96,102,108,114, \\ 120,126,132,138,144,150,156,162,168,174,180,186,192,198 \end{Bmatrix}$$

$$n(A) = 33$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{33}{200}$$

let B is the event that divisible by 8

$$B = \begin{Bmatrix} 8,16,24,32,40,48,56,64,72,80,88,96,104,112, \\ 120,128,136,144,152,160,168,176,184,192,200 \end{Bmatrix}$$

$$n(B) = 25$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{25}{200}$$

Here, A & B are not mutually exclusive event

$$A \cap B = \{24,48,72,96,120,144,168,192\}$$

$$n(A \cap B) = 8$$

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)} = \frac{8}{200}$$

Because A & B are not mutually exclusive event so we used

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = \frac{33}{200} + \frac{25}{200} - \frac{8}{200} = \frac{50}{200}$$

$$P(A \cup B) = \frac{1}{4}$$

Multiplication law of probability

If in two events used “and” that is A and B, then

we used Multiplication law.

There are two cases in Multiplication law, the

events are independent or dependent.

Independent events

A and B are said to be independent events, if

the occurrence and non-occurrence of event A

does not affects the events B.

In this case we used Multiplication law

$$P(A \cap B) = P(A) \times P(B)$$

Example 1: Two coins are rolled. Show that the event “head on the first coin” and the event “both face are same” are independent.

Solution: when two coins are tossed, the sample space is

$$S = \{HH, HT, TH, TT\}$$
$$n(S) = 4$$

let the event A head on first coin

$$A = \{HH, HT\}$$
$$n(A) = 2$$

Now the probability is

$$P(A) = \frac{n(A)}{n(S)}$$
$$P(A) = \frac{2}{4} = \frac{1}{2}$$

let B is the event both face are same

$$B = \{HH, TT\}$$

$$n(B) = 2$$

Now the probability is

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{2}{4} = \frac{1}{2}$$

Two event are independent so

$$P(A \cap B) = P(A) \times P(B)$$

Example 2: A beg contain 3 red candies and 7 blue candies. Two candies are taken from the bag with replacement. Find the probability the one candies is red and one candies is blue.

Solution: The total candies in beg or sample space is

$$S = \{R_c, R_c, R_c, B_c, B_c, B_c, B_c, B_c, B_c, B_c\}$$
$$n(S) = 10$$

let A denote the red candies

$$A = \{R_c, R_c, R_c\}$$
$$n(A) = 3$$

Probability red candies

$$P(A) = \frac{n(A)}{n(S)}$$
$$P(A) = \frac{3}{10}$$

let B denote the blue candies

$$B = \{B_c, B_c, B_c, B_c, B_c, B_c, B_c\}$$
$$n(B) = 7$$

Probability blue candies

$$P(B) = \frac{n(B)}{n(S)}$$
$$P(B) = \frac{7}{10}$$

Two event are independent so

$$P(A \cap B) = P(A) \times P(B)$$
$$P(A \cap B) = \frac{3}{10} \times \frac{7}{10}$$
$$P(A \cap B) = \frac{21}{100}$$

Dependent events

A and B are said to be dependent events, if the occurrence and non-occurrence of event A affected the event B.

In this case we used Multiplication law

$$P(A \cap B) = P(A) \times P\left({}^{B}\!/_{A}\right)$$

OR

$$P(A \cap B) = P(B) \times P\left({}^{A}\!/_{B}\right)$$

Example 2: A Beg has 4 white balls and 5 blue balls, we draw two balls from beg. Find the probability that first is white balls and second is blue balls.

Solution: The total balls in beg or sample space is

$$S = \{W_b, W_b, W_b, W_b, B_b, B_b, B_b, B_b, B_b\}$$
$$n(S) = 9$$

let A denote the white ball

$$A = \{W_b, W_b, W_b, W_b\}$$
$$n(A) = 4$$

Probability of white ball is

$$P(A) = \frac{n(A)}{n(S)}$$
$$P(A) = \frac{4}{9}$$

let B denote the blue ball

$$B = \{B_b, B_b, B_b, B_b, B_b\}$$
$$n(B) = 5$$

Probability blue ball when probability of white ball occurred

$$P\left({}^{B}/_{A}\right) = \frac{n(B)}{n(S)}$$
$$P\left({}^{B}/_{A}\right) = \frac{5}{8}$$

Two event are dependent so

$$P(A \cap B) = P(A) \times P\left({}^{B}/_{A}\right)$$
$$P(A \cap B) = \frac{4}{9} \times \frac{5}{8}$$
$$P(A \cap B) = \frac{5}{18}$$

Example 2: A Beg contains 15 items, 4 of which are defective and 11 are good. Two item are selected, what is the probability that the first is good and the second is defective.

Solution: The total items in beg or sample space is

$$S = \{D, D, D, D, G, G, G, G, G, G, G, G, G, G, G\}$$

$$n(S) = 15$$

let A denote the item is good

$$A = \{G, G, G, G, G, G, G, G, G, G, G\}$$

$$n(A) = 11$$

Probability of good item is

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A) = \frac{11}{15}$$

let B denote the defective items

$$B = \{D, D, D, D\}$$
$$n(B) = 4$$

Probability of defective item when probability of good items occurred

$$P\left({}^{B}/_{A}\right) = \frac{n(B)}{n(S)}$$
$$P\left({}^{B}/_{A}\right) = \frac{4}{14}$$

Two event are dependent so

$$P(A \cap B) = P(A) \times P\left({}^{B}/_{A}\right)$$
$$P(A \cap B) = \frac{11}{15} \times \frac{4}{14}$$
$$P(A \cap B) = \frac{22}{105}$$

Conditional Probability

The sample space for an experiment must be changed when some additional information pertaining to the outcome of the experiment is received. The effect of such information is to reduce the sample space by excluding some outcomes as being impossible which before receiving the information where believed possible. The probability associated with such a reduced sample space is called Conditional Probability.

If A and B are two events in a sample space S, if P(A) is not equal to zero then the conditional probability of the event A given that event B has occurred as

$$P\left(A/_{B}\right) = \frac{P(A \cap B)}{P(B)}$$

OR

$$P\left(B/_{A}\right) = \frac{P(A \cap B)}{P(A)}$$

Example 1: Two coins are tossed. Find the probability that both are heads given that at least one is head.

Solution: The sample space of two coins is

$$S = \{HH, HT, TH, TT\}$$

$$n(S) = 4$$

let A denote the event both are heads, and B denote the event at least one is head and

$$A = \{HH\}$$

$$n(A) = 1$$

$$B = \{HH, HT, TH\}$$

$$n(B) = 3$$

$$A \cap B = \{HH\}$$

$$n(A \cap B) = 1$$

Probability at least one head is

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{3}{4}$$

now common probability

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)}$$

$$P(A \cap B) = \frac{1}{4}$$

Conditional probability is

$$P\left({}^{A}/_{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$P\left({}^{A}/_{B}\right) = \frac{\frac{1}{4}}{\frac{3}{4}}$$

$$P\left({}^{A}/_{B}\right) = \frac{1}{3}$$

Example 2: A fair dice is rolled. Find the probability that the face is even given that the face is less than 4.

Solution: The sample space of a one dice is

$$S = \{1,2,3,4,5,6\}$$
$$n(S) = 6$$

let A denote the even face, and B denote the event that the face is less than 4

$$A = \{2,4,6\}$$
$$n(A) = 3$$
$$B = \{1,2,3\}$$
$$n(B) = 3$$
$$A \cap B = \{2\}$$
$$n(A \cap B) = 1$$

Probability that the face is less than 4

$$P(B) = \frac{n(B)}{n(S)}$$

$$P(B) = \frac{3}{6} = \frac{1}{2}$$

now common probability is

$$P(A \cap B) = \frac{n(A \cap B)}{n(S)}$$

$$P(A \cap B) = \frac{1}{6}$$

Conditional probability is

$$P\left({}^{A}/_{B}\right) = \frac{P(A \cap B)}{P(B)}$$

$$P\left({}^{A}/_{B}\right) = \frac{\frac{1}{6}}{\frac{1}{2}}$$

$$P\left({}^{A}/_{B}\right) = \frac{1}{3}$$